

THERMABRICK · TEST REPORT

ThermaBrick test article test report

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DOCUMENT CONTROL

Document ID	TBK-TST-002	Version	0.2
Status	Draft	Date	2026-09-24
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/thermabrick		
Source	docs/05-tests/TBK-TST-002-test-report.md @ a07b0c2		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-24	Amish Chadha	Kit 1.1.4: front matter adds environment: lab
0.1	2026-09-24	Amish Chadha	Template. Structure follows TBK-TST-001; sections 5, 6 and 8 and the summary rows for TP3 to TP5 are filled from the analysis results by tbk_tst_002_fill.py

ThermaBrick test article test report

Summary to be written once the tests are complete. State in two or three sentences whether the test article met TBK-TST-001, give the measured sand conductivity, contact conductance and insulation conductivity against the values assumed in TBK-CAL-001, and say what they mean for the full-scale charge-time requirement R3 (section 8).

Procedure	Dates	Pass criterion (TBK-TST-001)	Result	Pass
TP0 Inspection and instrument checks		Channels within ± 2 K at 0 °C and 100 °C		
TP1 Limit and fault response		Trip at 600 °C $\pm$ 10 K; latches; firmware safe state within 10 s		
TP2 Bake-out and commissioning		1 M Ω or more per heater at 500 V; no smoke in the last 12 h		

| TP3 Charge | | Fitted RMS 10 K or less; time to 80 % within 10 %; power correction within 5 % | *filled by script* | |

TP4 Cool-down | | Loss at 450 °C within 20 % of 249 W before fitting | *filled by script* | | | TP5 Discharge | | Hourly duty within 15 % of the model, no refit | *filled by script* | |

| TP6 Cycling | | Circumference growth under 0.25 % after 25 cycles | | |

Table 1. Results against TBK-TST-001. The TP3 to TP5 rows are filled by `tbk_tst_002_fill.py` ; enter the dates by hand.

■ 1. Test article as built

The test article was built to TBK-PRC-002 and `docs/07-build/test-article-build.md` , with the deviations in Table 2. Each deviation comes from the build log, and each should have its document change committed before this report is released.

Deviation	Build log entry	Effect on the tests	Document changed
None recorded yet			

Table 2. Deviations from the design.

Item	Value
Sand mass and depth	kg, mm (build stage F)
Heater cold resistances H1 to H4	Ω
Firmware commit	
Analysis script commit	
Parts cost, actual against the \$727.70 BOM	\$

Table 3. As-built record.

■ 2. Instrumentation and calibration (TP0)

Channel	Reading at 0 °C	Reading at 100 °C (corrected for elevation)	Offset applied
T1			
T2 (REX-C100)			
T3			
T4			
T5			
T6			

Table 4. Thermocouple checks.

Blower duty	Airflow, mean of three fills (L/s)
25 %	
50 %	
75 %	
100 %	

Table 5. Airflow calibration, as loaded into the firmware with `/config?air=`.

■ 3. TP1: Limit and fault response

Trigger	Expected	Measured	Pass
REX-C100 trip on T2 (reference probe reading)	600 °C ± 10 K		
Latch holds after T2 cools	Stays open until START		
STOP with K1 latched	Drops		
Power loss and return	Stays open until START		
T1 unplugged	Heaters off within 10 s		
T1 above 575 °C (test build)	Heaters off within 10 s		
ESP32 reset	Heaters off within 10 s		
Logging host stopped	Heaters off within 70 s		

Table 6. Limit and fault results.

■ 4. TP2: Bake-out and commissioning

Heater	Insulation resistance before (M Ω at 500 V)	After (M Ω at 500 V)
H1		
H2		
H3		
H4		

Table 7. Heater insulation resistance.

Record when smoke and odor stopped, the time taken to settle at 150 °C, and any condensation or steam at the leg openings.

■ 5. TP3 and TP4: Fitted properties

Filled by `tbk_tst_002_fill.py` from `results/tbk_tst_001_fit.json`.

■ 6. TP5: Discharge prediction

Filled by `tbk_tst_002_fill.py` from `results/tbk_tst_001_tp5.json`.

Record the peak T5, the stack exit temperature from the contact thermometer, and whether the blower reached 4.0 L/s hot.

■ 7. TP6: Cycling

Cycle	Circumference at 100 mm (mm)	At 330 mm (mm)	At 560 mm (mm)	Time for T1 to reach 550 °C (h)
0				
5				
10				
15				
20				
25				

Table 8. Drum circumference and charge behavior over 25 cycles.

State the growth at each height after 25 cycles, the trend extrapolated to 1,000 cycles (full-scale R19), the brick condition, and the heater resistances after cycling.

■ 8. Implications for the full-scale design

Filled by `tbk_tst_002_fill.py` : TBK-CAL-001 re-run with the fitted values.

State whether TBK-CAL-001 should be reissued with the fitted values, and which option in TBK-PRC-001, section 8 (charge time), the results support.

■ 9. Anomalies and observations

List anything unexpected, with the build log entry and data file where it is recorded.

■ 10. Conclusions and recommendations

Numbered conclusions, each traced to a table or figure above, then the recommended next steps.

■ Appendix A. Data files

Procedure	File	Firmware
	docs/05-tests/data/	

Table 9. Raw data files. These are never edited; corrections are applied in the analysis script.